

Figure 1: Dose response relationship that is completely mediated by exposure, with PK difference between adult and pediatric

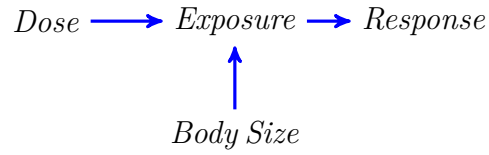


Figure 2: Full extrapolation with allometrically scaled PK

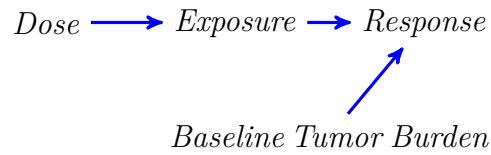


Figure 3: Trastuzumab Naive

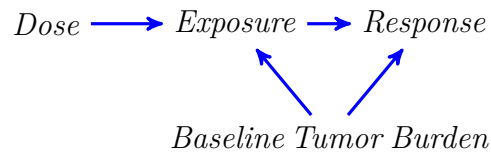


Figure 4: Trastuzumab

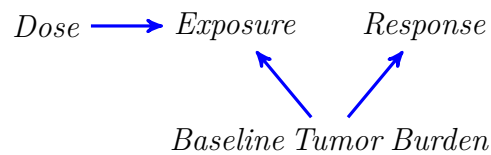


Figure 5: Trastuzumab post hoc

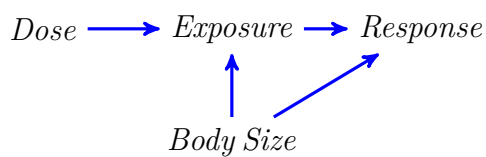


Figure 6: SGLT-2

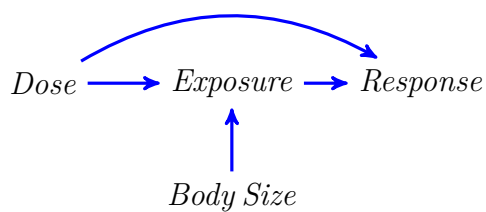


Figure 7: Questions before full extrapolation with allometric scaling

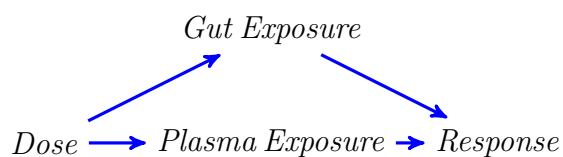


Figure 8: Show mediator?



Figure 9: Or not?

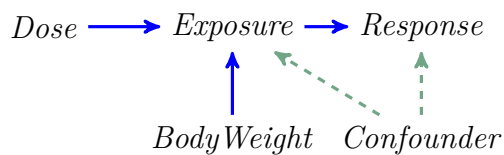


Figure 10: More questions before full extrapolation with allometric scaling

$$\begin{array}{l}
 D \perp\!\!\!\perp R^{(d)} \mid E \\
 D \not\perp\!\!\!\perp R^{(d)} \mid E
 \end{array}$$

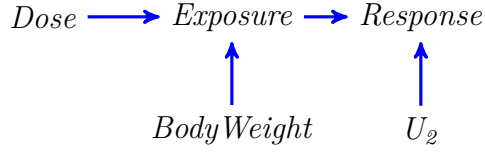


Figure 11: Working DAG within adults and within peds

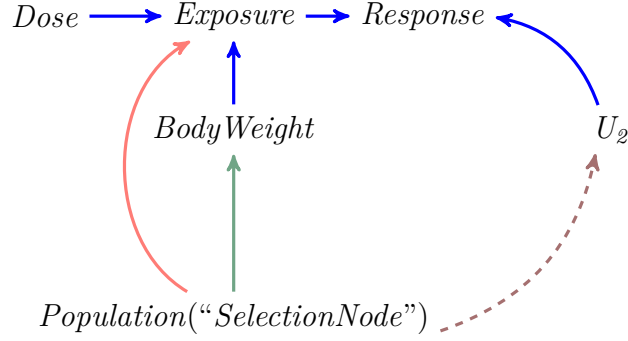


Figure 12: Selection graph for full extrapolation: justification

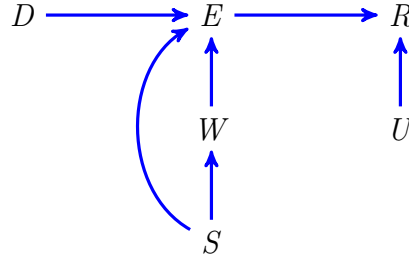


Figure 13: Working selection diagram for full extrapolation.

$D$  = dose or treatment status  
 $E$  = exposure (summary metric)  
 $R$  = Response  
 $W$  = Body weight  
 $U$  = Unmeasured effect modifier  
 $S$  = Selection node (pediatric status)

|                | Adult                            | Pediatric                          |
|----------------|----------------------------------|------------------------------------|
| Observational  | $P(\mathbf{v} D = d)$            | $P^*(\mathbf{v} D = d)$            |
| Interventional | $P(\mathbf{v} \text{do}(D = d))$ | $P^*(\mathbf{v} \text{do}(D = d))$ |

$$P^*(r|\text{do}(D = d))$$

$$P(r \mid \text{do}(D = d)) = \int_e P^*(r \mid e, \text{do}(D = d)) P^*(e \mid \text{do}(D = d)) \quad (1)$$

$$= \int_e P^*(r \mid e) P^*(e \mid \text{do}(D = d)) \quad (2)$$

$$\begin{aligned} P^*(r \mid \text{do}(D = d)) &= \int_e P^*(r \mid e, \text{do}(D = d)) P^*(e \mid \text{do}(D = d)) de && \text{(Law of total prb.)} \\ &= \int_e P^*(r \mid e) P^*(e \mid \text{do}(D = d)) de && (D \perp R \mid E) \\ &= \int_e P(r \mid e) P^*(e \mid \text{do}(D = d)) de && (S \perp R \mid E) \end{aligned}$$

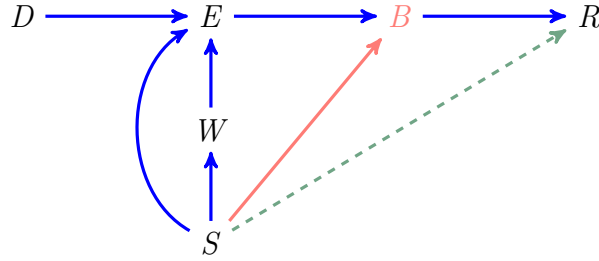


Figure 14: Working selection diagram for full bridging biomarker.

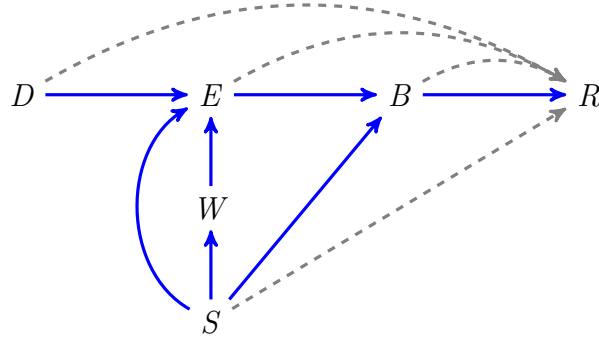


Figure 15: Bridging biomarker selection diagram that could be wrong

$$\begin{aligned} P^*(r \mid \text{do}(D = d)) &= \int_b P^*(r \mid b, \text{do}(D = d)) P^*(b \mid \text{do}(D = d)) db && \text{(Law of total prb.)} \\ &= \int_b P^*(r \mid b) P^*(b \mid \text{do}(D = d)) db && (D \perp R \mid B) \\ &= \int_b P(r \mid b) P^*(b \mid \text{do}(D = d)) db && (S \perp R \mid B) \end{aligned}$$

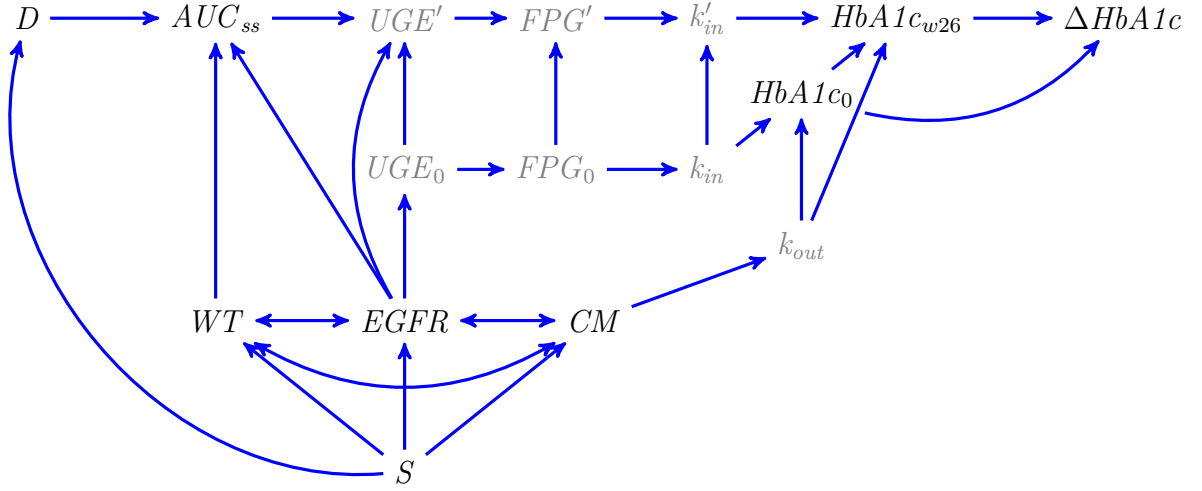


Figure 16: DINAMO Selection Diagram

- $D$  = dose or treatment status  
 $AUC_{ss}$  = exposure at PK steady-state  
 $UGE_0, UGE'$  = urinary glucose excretion, pre- and post-treatment  
 $FPG_0, FPG'$  = fasting plasma glucose, pre- and post-treatment  
 $k_{in}, k'_{in}$  = rate constant for HbA1c synthesis, pre- and post-treatment  
 $k_{out}$  = rate constant for HbA1c degradation / elimination  
 $HbA1c_0, HbA1c_{w26}$  = hemoglobin A1c at baseline and week 26  
 $\Delta HbA1c$  = Change from baseline in hemoglobin A1c  
 $WT$  = Body weight  
 $EGFR$  = estimated glomerular filtration rate  
 $CM$  = Concomitant medication usage  
 $S$  = Selection node (pediatric status)

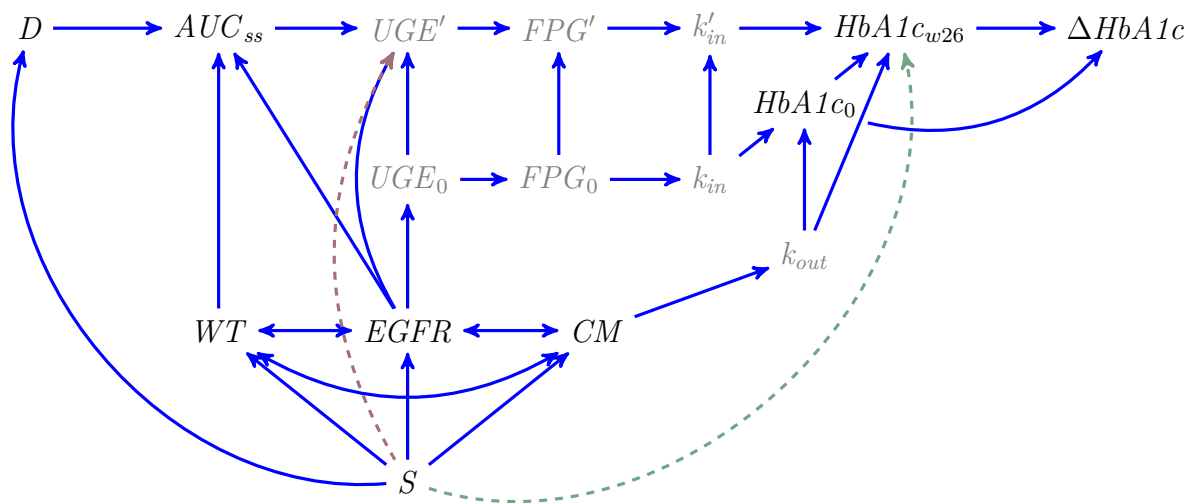


Figure 17: DINAMO Selection Diagram

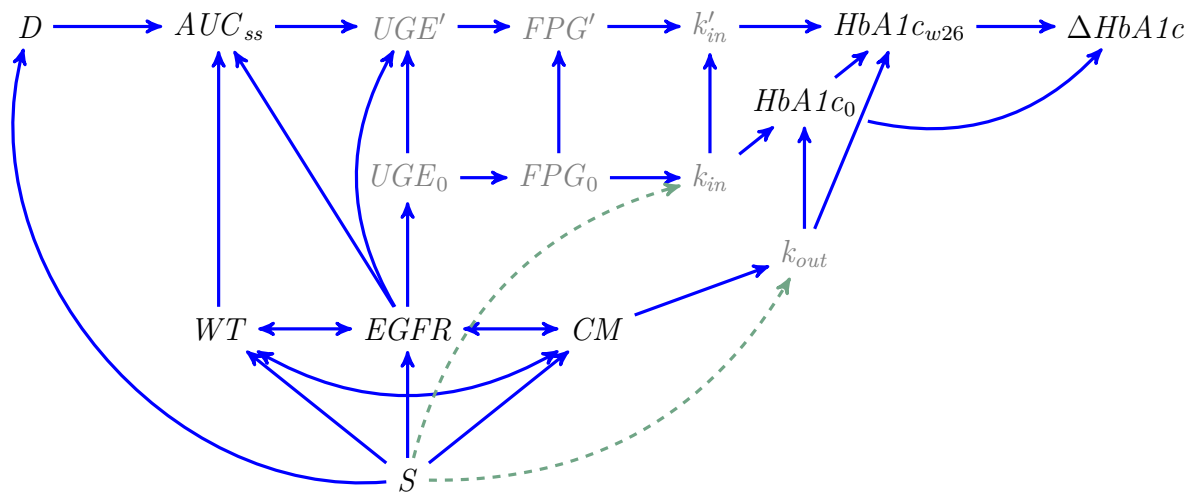


Figure 18: DINAMO Selection Diagram

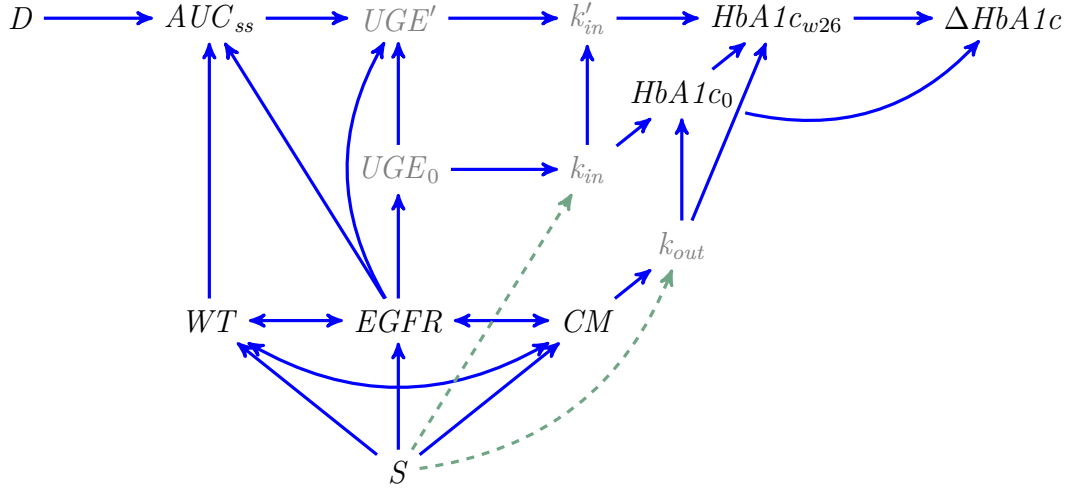


Figure 19: DINAMO Threat 2

$$E^* \left[ \Delta HbA1c^{(D=d)} - \Delta HbA1c^{(D=0)} \right]$$

$$E^* \left[ \Delta HbA1c^{(D=d)} \right]$$

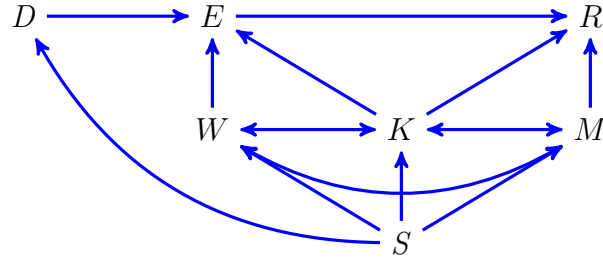


Figure 20: Abbreviated DINAMO selection diagram

- $D$  = dose or treatment status
- $E$  = exposure at PK steady-state
- $R$  = Change from baseline in hemoglobin A1c
- $W$  = Body weight
- $K$  = Kidney function (EGFR)
- $M$  = Concomitant medication usage

$$\begin{aligned}
 P^*(r \mid \text{do}(d)) &= \int_w \int_k \int_m P^*(r \mid \text{do}(d), w, k, m) P^*(w, k, m \mid \text{do}(d)) dm dk dw && \text{(Law of t.p.)} \\
 &= \int_w \int_k \int_m P^*(r \mid \text{do}(d), w, k, m) P^*(w, k, m) dm dk dw && ((W, K, M) \perp D)_{G_{\overline{D}}}
 \end{aligned}$$

$$\begin{aligned}
&= \int_e P^*(r \mid \text{do}(d), e, w, k, m) P^*(e \mid w, k, m, \text{do}(d)) de && \text{(Law of t.p.)} \\
&= \int_e P(r \mid \text{do}(d), e, w, k, m) P^*(e \mid w, k, m, \text{do}(d)) de && (R \perp S \mid W, K, M)_{G_{\overline{D}}} \\
&= \int_e P(r \mid \text{do}(d), e, w, k, m) P^*(e \mid w, k, \text{do}(d)) de && (E \perp M \mid W, K)_{G_{\overline{D}}} \\
&= \int_e P(r \mid e, k, m) P^*(e \mid w, k, \text{do}(d)) de && (R \perp (W, D) \mid W, K, M, E)_{G_{\overline{D}}}
\end{aligned}$$

$$P^*(r \mid \text{do}(d)) = \int_w \int_k \int_m \int_e P(r \mid e, k, m) P^*(e \mid w, k) P^*(w, k, m) dm dk dw de$$

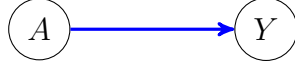
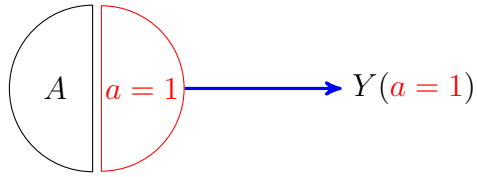
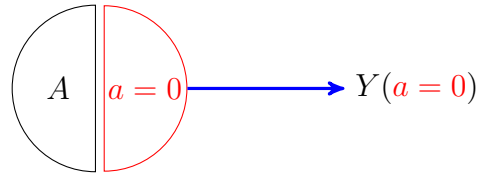


Figure 21: Simplest DAG



(a) SWIG for Experimental Treatment



(b) SWIG for Reference Treatment

Figure 22: Two possible SWIGs for DAG in Figure 2

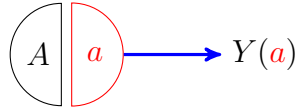


Figure 23: General graph representing the two SWIGs in Figure 3

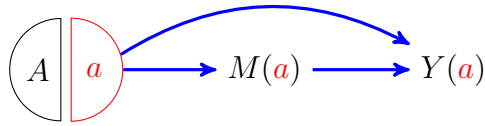


Figure 24: SWIG for an ITT Estimand

$$\begin{aligned}
\Delta_{ITT} &= E[Y(1)] - E[Y(0)] \\
&= E[Y(1) \mid A = 1] - E[Y(0) \mid A = 0] && (Y(a) \perp\!\!\!\perp A) \\
&= E[Y \mid A = 1] - E[Y \mid A = 0] && (\text{consistency})
\end{aligned}$$

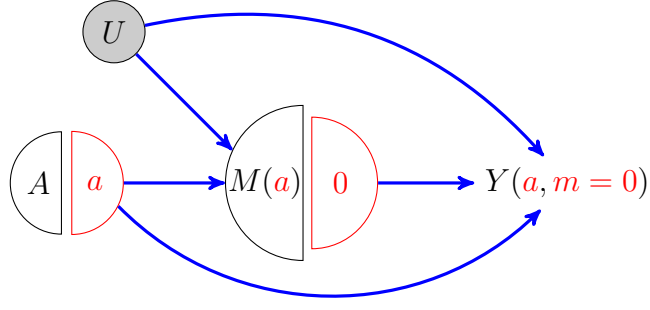


Figure 25: SWIG for a hypothetical estimand with unobserved confounding

$$\begin{aligned}
 \Delta_{hypo} &= E[Y(1, m)] - E[Y(0, m)] \\
 &= E[Y(1, m)|A = 1] - E[Y(0, m)|A = 0] \quad (Y(a, m) \perp\!\!\!\perp A)
 \end{aligned}$$

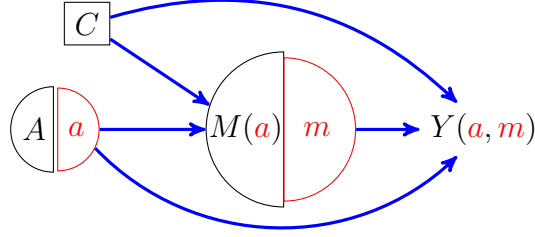


Figure 26: SWIG for a hypothetical estimand with no unobserved confounding

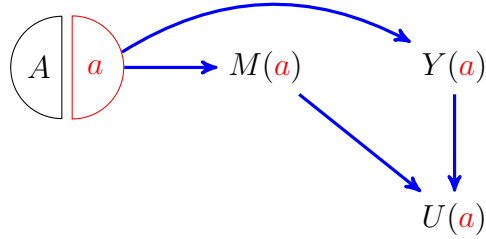


Figure 27: SWIG for a composite estimand

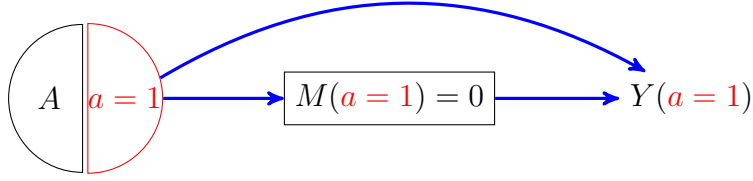


Figure 28: SWIG where active treatment is given for those who would not have the IE under treatment (i.e. conditioning on  $M(a=1)=0$ )

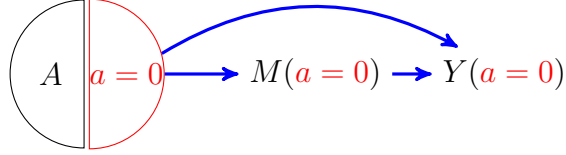


Figure 29: SWIG where reference treatment is given for those who would not have the IE under treatment  $M(a=1)=0$

- $M_1$  = Intake of short acting pain relief medication
- $M_2$  = Treatment discontinuation due to Adverse Event, Loss of Efficacy, or intake of prohibited medications
- $M_3$  = Change of dose of allowed concomitant medication for pain
- $M_4$  = Treatment discontinuation due to Administrative or Other reasons

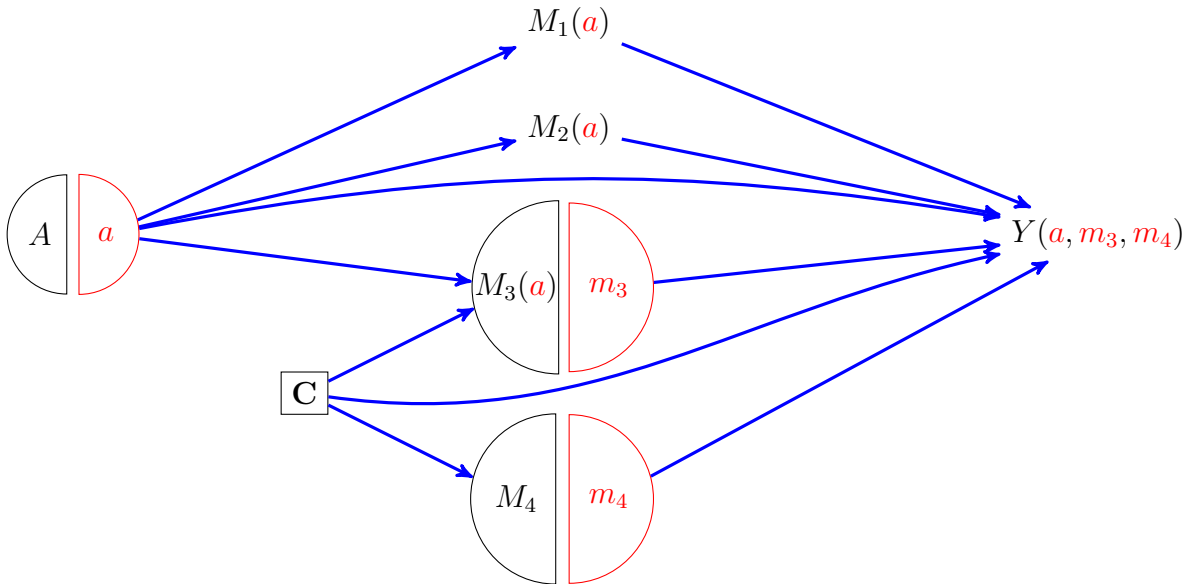


Figure 30: SWIG for a Clinical Trial in Chronic Pain